Homework

HW7 — An evolution with variable input — 60 min.

We look for a function y defined and differentiable on $[0;+\infty[$ such that $y'(t) = -y(t) + 2t + 1$ and $y(0) = 3$.

1. Find two real numbers a and b such that $g(t) = at + b$ is a solution of the differential equation.
2. Deduce the expression of the solution that satisfies $y(0) = 3$.
3. Study the variations of this solution and state its minimum and the instant at which it is attained.
4. Write a Python function with argument N, a strictly positive integer, that applies Euler's method on $[0;2]$ with step $\frac{2}{N}$ and returns the approximation of $y(2)$. Compare with the exact value for $N = 10$, then $N = 100$.
5. Why is checking the differential equality at only a few instants not enough to prove that a function is a solution?